Technical Report: Quantification of Credit Risk

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1. Introduction and Data Description

The dataset was collected in 2005 from a large bank in Taiwan that operated as both a cash and credit card issuer. It contains 30,000 credit card holders as instances with 23 features involving demographic information, credit limit, and monthly payment history. The dataset contains no missing values, therefore, it does not require imputation prior to analysis.

2. Inferential Framework

For our inference question, we ask "can a client's repayment history significantly affect the probability of defaulting next month?". The population of interest are the credit card holders of the Taiwanese bank in 2005. The parameter of interest is the logistic regression coefficient β associated with repayment delay status (PAY_0) that represents the change in log-odds of default per unit increase in payment delay.

In contrast to prediction, we pose the question of "will clients default on their next payment given their current financial profile?". The target variable is default_payment_next_month, which is a binary outcome where 1 indicates default and 0 indicates no default. Model performance is evaluated with binary cross entropy as the loss metric with Precision-Recall AUC (PR-AUC) used as the primary evaluation criterion due to the class imbalance in the dataset.

The distinction between inference and prediction is that inference's goal is to find β for PAY_0 and prediction's goal is to minimize the misclassification of default labeling. Their approach to their goal is significantly different too. Inference focuses on being unbiased and interpretable, meanwhile prediction aims to be generalizable by achieving a fair AUC score. The output in the end will be a coefficient or confidence interval for inference and binary label for prediction.

3. Likelihood and Maximum Likelihood Estimation

The logistic regression formula was the choice for modeling of the binary nature of the project outcome that follows the Bernoulli distribution. Compared to linear regression, the logistic model ensures that the predictions are limited to the interval (0, 1) through the sigmoid function, therefore, fitting with the probability interpretation that linear regression cannot provide with its unbounded interval. Given the x features in the dataset, the probability of default is modeled as:

$$P(Y = 1 \mid \mathbf{x}) = \sigma(\mathbf{x}^T\boldsymbol{\beta}) = \frac{1}{1 + e^{-\mathbf{x}^T\boldsymbol{\beta}}}$$

With 29,965 independent observations, each client's outcome within 0 and 1 follows the Bernoulli distribution with success probability $p_i = \sigma(x_i^T\beta)$. The joint likelihood function looks like:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} \sigma(\mathbf{x}_i^T\boldsymbol{\beta})^{y_i} \left(1 - \sigma(\mathbf{x}_i^T\boldsymbol{\beta})\right)^{1-y_i}$$

After taking the log, it will help optimize and give a log-likelihood function:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[y_i \log \sigma(\mathbf{x}_i^T\boldsymbol{\beta}) + (1 - y_i) \log \left(1 - \sigma(\mathbf{x}_i^T\boldsymbol{\beta})\right)\right]$$

Because there is no closed-form solution, the MLE is formulated computationally with the Newton-Raphson method that updates the coefficient estimate with each iteration using the gradient and Hessian of the log-likelihood. The process converged after 7 iterations to confirm that the log-likelihood surface is strictly concave and the MLE is unique.

The full model resulted in a log-likelihood of -13,924 compared to the intercept-only null model of -15,836. This means the 23 features were able to improve the predictive power. McFadden's Pseudo R^2 of 0.121 suggests that the model can reduce the baseline uncertainty by 12%, which is satisfactory for a behavioral financial dataset.

The MLE for PAY_0 was discovered to be β = 0.6497 with a 95% confidence interval of [0.611, 0.689], which is the most significant coefficient in the model. It corresponds to a 92% ($e^{0.6497} \approx 1.92$) increase in odds of default per unit increase in payment delay. PAY_0 is treated as continuous here, though it is technically ordinal. We acknowledge this as a limitation, however, the effect remains highly significant regardless of coding choice. Features, including PAY_4 to PAY_6 and BILL_AMT2 to BILL_AMT6, are less statistically significant due to its lag in time, implying more recent behavior has more influence over the credit risk evaluation. A log-likelihood surface assessment was conducted over an interval of values for β = PAY_0 to confirm a single, unique peak at the MLE. The inverse Fisher information matrix also shows that the confidence interval is entirely above 0.

4. Frequentist Inference

To quantify the uncertainty around our estimate of β for PAY_0, we conducted a frequentist analysis based on the asymptotic properties of the maximum likelihood estimator. Given our large sample size of n = 29,965 and 23 features included in the model, the Central Limit Theorem guarantees that $\hat{\beta}$ is approximately normally distributed around the true parameter β. We verified this empirically by drawing

B = 1,000 bootstrap samples with replacement from the dataset and refitting the full multivariate logistic regression at each iteration. The resulting sampling distribution of $\hat{\beta}$ for PAY_0 was approximately symmetric and bell-shaped, with a mean of 0.6495 and standard deviation of 0.0232, consistent with the asymptotic normality predicted by the CLT.

We then constructed two 95% confidence intervals for β using different methods. The theory-based interval, computed from the inverse Fisher information matrix as $\hat{\beta} \pm 1.96 \cdot SE$, yielded [0.6107, 0.6887] with point estimate $\hat{\beta} = 0.6497$ and standard error SE = 0.0199, which is consistent with the MLE results reported in Section 3. The percentile bootstrap interval, taken as the 2.5th and 97.5th percentiles of the bootstrap distribution, yielded [0.6042, 0.6927]. The bootstrap interval is slightly wider than the theory-based interval, which is expected given that bootstrap relies on fewer parametric assumptions. The close agreement between the two confirms that the normal approximation is reliable in this setting. Importantly, neither interval contains zero, providing initial evidence that β is meaningfully different from zero.

We formalized this conclusion through a Wald hypothesis test of H_0: $\beta = 0$ against H_1: $\beta \neq 0$. The z-statistic, computed as $z = \hat{\beta}/SE = 32.65$, places our estimate over 32 standard deviations from zero, with a corresponding p-value indistinguishable from zero at standard floating-point precision. We therefore reject the null hypothesis at any conventional significance level. This conclusion is consistent with the confidence interval result, since the 95% CI excludes zero.

Substantively, these results indicate that repayment delay status (PAY_0) is a statistically significant predictor of default at the population level, even after controlling for all other demographic and financial features. The estimated coefficient $\hat{\beta} = 0.6497$ corresponds to an odds multiplier of approximately $e^{0.6497} \approx 1.92$, meaning that a one-unit increase in PAY_0 roughly doubles the odds of default, which aligns with the interpretation provided in Section 3. The percentile bootstrap interval is sometimes interpreted in a manner closer to the Bayesian credible interval, whereas the strict frequentist interpretation is that 95% of intervals constructed by this procedure across repeated samples would contain the true β.

5. Bayesian Analysis

Bayesian's approach is to find the optimal parameter value based on the updated posterior distribution of potential parameter values from updated knowledge about a phenomenon. Unlike the frequentist approach mentioned earlier, rather than treating β as a fixed unknown variable to be discovered, Bayesian estimation treats β as a random variable with a probability distribution. The goal for Bayesian's method here would be to characterize the full range of plausible values for β given the data and prior knowledge all together.

Here we adopt the Baye's theorem, which combines a prior distribution over β with the logistic regression likelihood to produce a posterior distribution:

$$P(\beta \mid data) \propto P(data \mid \beta) \cdot P(\beta)$$

For the prior, we placed an independent normal distribution on each of the 23 coefficients from the dataset (with a mean 0 and a standard deviation 10) expressed as $P(\beta) = N(0,\ 10)$. The use of a weakly informative prior (a prior that expresses no strong belief about the direction or magnitude of any coefficient before seeing the data) is intended so that we initialize the 'knowledge' about this distribution with a fair parameterized distribution. Nevertheless, given that we have about 30k of observations, the posterior distribution will inevitably be driven by data rather than what we set for the prior distribution.

While we did consider using grid search to search over all of the parameter combinations to find a plausible range for β, it is still computationally infeasible to search over all 23 features' combinations all together. Such as even with just sampling 10 candidate values per feature, the number of combinations will be 10^{23}, which is impossible for the machine to calculate manually over all combinations. To address this limitation, we adopt a methodology called the Laplace Approximation. The Laplace Approximation works by first finding the peak of the posterior distribution, called the Maximum A Posteriori (MAP) estimate, by finding the β that maximizes the sum of the log-likelihood and log prior:

$$\beta_{MAP} = argmax[l(\beta) + log\ P(\beta)].$$

This estimation is done by having each candidate β value to be scored by how well it fits the dataset and how consistent it is with the prior, in which the peak of the distribution – the best scoring estimate, will become our point estimate value.

Secondly, it approximates the shape of the posterior around that the best estimate value (or the peak of the distribution) by describing how it falls around the peak using Hessian Matrix H. The Hessian matrix measures the curvature of the log-posterior at the MAP value. In which a steep and narrow peak will produce a small covariance H^{-1} value, meaning there is a small range of uncertainty. While a flat, wide peak will produce a large covariance score, representing greater uncertainty range. The posterior is then approximated as a normal distribution centered at the MAP value with covariance H^{-1}, from which we extract the credible intervals for each feature in the model.

Overall, our experiments show that the MAP estimated a PAY_0 β value of 0.6497, with a 95% credible interval of [0.6107, 0.6887]. This result is similar to the results produced earlier with MLE's point estimate of 0.6497 as well as the frequentist's bootstrap confidence interval of [0.6042, 0.6927]. The 95% credible interval here can be interpreted as given the observed data and our prior, there is a 95% probability that the true value of β_{PAY_0} lies within this range.

6. Resampling and Simulation

To quantify uncertainty in our parameter estimate $\hat{\theta} = 0.2213$ (the sample default rate), we implemented bootstrap resampling with B = 10,000 iterations, drawing samples with replacement. The bootstrap procedure yielded a mean of 0.2213 (matching our point estimate) and standard error of 0.002379, remarkably close to the theoretical normal approximation SE = 0.002398. This near-perfect agreement demonstrates bootstrap's distribution-free approach. Using the percentile method, we constructed a 95% confidence interval [0.2166, 0.2259], meaning that if we repeatedly sampled from the population and constructed intervals using this bootstrap procedure, approximately 95% of those intervals would contain the true population default rate. The narrow interval width reflects high precision from our large sample size.

We validated bootstrap accuracy through Monte Carlo simulation by drawing M = 1,000 samples from a Bernoulli distribution and computing sample proportions for each. The Monte Carlo standard error of 0.002418 matched our bootstrap estimate within 0.00004. All three approaches (bootstrap, Monte Carlo, and theoretical) agree on the standard error within a negligible margin, validating the robustness of our inference procedures.

The small standard error (0.002379) indicates our estimate would typically vary by only ±0.24 percentage points across repeated samples. This low variability has contrasting implications for inference versus prediction. For inference, low variance is desirable because it ensures stable parameter estimates and narrow confidence intervals, allowing precise statements about the population default rate. The consistency between bootstrapping and Monte Carlo validates our inference framework despite the binary data. However, these resampling methods quantify sampling variability only, assuming our sample is representative. Systematic biases in data collection are not captured by bootstrap-based uncertainty quantification. For prediction, low variance is necessary but insufficient; we also need low bias for accurate future predictions. The high precision of our parameter estimate does not automatically translate to strong predictive performance, particularly given the class imbalance that biases classifiers toward the majority class.

7. Predictive Modeling

We implemented a stratified 70/30 train-test split (n = 20,975 training, n = 8,990 test) preserving the 22.13% default rate in both partitions. Five-fold cross-validation confirmed model stability with test accuracy of 0.8091 ± 0.0051 and negligible train-test gaps (< 0.003), demonstrating consistent generalization. We fitted baseline and class-balanced logistic regression classifiers on standardized

features, evaluating performance using Precision-Recall AUC (PR-AUC) rather than ROC-AUC because PR-AUC better captures minority class discrimination in severely imbalanced datasets.

The baseline model achieved PR-AUC of 0.51 on the test set. To contextualize this score: a random classifier predicting defaults with 22% probability (matching class prevalence) achieves PR-AUC = 0.22, while perfect classification yields 1.0. Our 0.51 represents meaningful improvement (2.3× better than random) but reveals substantial room for enhancement. The model exhibited high bias and low variance: recall was only 24.1%, missing 76% of actual defaults due to conservative predictions, while tight CV confidence bands (std < 0.005) and minimal train-test gaps confirmed stable generalization without overfitting. This reflects the fundamental tradeoff: the model's simplicity ensures consistent predictions (low variance) but causes systematic underdetection of defaults (high bias). When we implemented class-balance, recall improved from 24% to 64% (+40pp) while precision dropped from 73% to 38%, achieving PR-AUC = 0.50. Both models demonstrated similar discrimination capability but operated at different precision-recall operating points, with the class-balanced version detecting 2.7× more defaults.

We also implemented double bootstrapping with $B_1 = 100$ outer bootstrap iterations and $B_2 = 5$ inner cross-validation folds that validated our model selection by separating training from evaluation. Results showed baseline PR-AUC = 0.5070 ± 0.0017 and class-balanced PR-AUC = 0.5034 ± 0.0013, with tight confidence intervals proving these moderate scores represent genuine, reproducible performance. The averaged PR curves across 100 iterations were nearly identical to our original single split, with minimal variation (std < 0.002) confirming stability. Credit risk prioritizes default detection, so we selected the class-balanced model. It achieved 63.65% ± 0.96% recall versus the baseline's 24.11% ± 1.11%, a significant improvement that justifies the precision tradeoff.

8. Bias–Variance / Regularization and Shrinkage

To examine the bias-variance tradeoff, we used a grid search on an unpenalized logistic regression baseline against Ridge (L2), Lasso (L1), and Elastic Net, which were tuned via five-fold stratified cross-validation over PR-AUC on the training set. All features were standardized prior to fitting, and cross-validation searched over 50 candidate values of C on a logarithmic grid.

The optimal regularization strengths differed meaningfully across methods: Ridge selected C = 0.0268, reflecting relatively low penalization, while Lasso selected C = 0.0072 ($\lambda \approx 138.9$) and Elastic Net selected C = 0.0052, with l1_ratio = 0.8, both of which reflected stronger shrinkage. Despite this, all four models achieved nearly identical PR-AUC scores of approximately 0.501 on the test set. Lasso and Elastic Net each compressed 7 of 23 coefficients to near-zero, while Ridge compressed only 1, yet neither

sparsity nor stronger penalization translated into any measurable predictive gain over the unpenalized baseline. This confirms the dataset operates in a low-variance regime. With nearly 30,000 observations and only 23 features, the MLE is already a stable estimator, and the primary effect of adding a strong penalty is to introduce unnecessary bias without recovering variance.

We additionally applied James–Stein shrinkage to group-level default rates stratified by PAY_0, working in the arcsin-square-root variance-stabilized space. The estimated shrinkage factor was 0.108, indicating mild adjustment. On this test split, MLE achieved a marginally lower MSE (0.14353) than James–Stein (0.14395). This finite-sample reversal is unsurprising: PAY_0 groups are highly unequal in size and heterogeneous in default rates, ranging from 0.13 to 0.77, meaning shrinkage toward the grand mean introduces more bias than it removes variance.

Taken together, both analyses confirm that this dataset does not benefit from shrinkage-based regularization.

9. Inference vs Prediction Reflection

This project shows that inference and prediction are related, but they are ultimately useful for different purposes. Inference helped us understand why default risk changes, while prediction helped us decide which clients are most likely to default next month. In our inferential analysis, repayment history–especially PAY_0–was clearly important. The estimated coefficient for PAY_0 was positive and statistically significant, with a 95% confidence interval entirely above zero, showing that increased repayment delay is associated with a higher probability of default. This makes inference valuable because it provides an interpretable explanation of credit risk and identifies which factors matter most at the population level.

However, for the actual credit-risk problem, prediction is more useful. A bank ultimately needs a model that can classify future borrowers as risky or not risky, since lending decisions depend on future outcomes rather than coefficient interpretation alone. Our predictive results show why this distinction matters. The baseline logistic model produced a PR-AUC of 0.51, which is meaningfully better than random but still leaves substantial room for improvement. More importantly, its recall was only about 24%, meaning it missed most actual defaults. After class balancing, recall increased to about 64%, even though precision fell. This tradeoff makes prediction the more practically relevant framework here, because in credit risk it is often more important to detect likely defaulters than to preserve a highly conservative classifier.

Uncertainty affected the two frameworks differently. For inference, uncertainty was expressed through confidence intervals, bootstrap results, and Monte Carlo validation, which showed that the estimated default rate and the effect of repayment history were stable and precise. For prediction,

uncertainty appeared through cross-validation and double bootstrapping, indicating that the model's performance was consistent across resamples, even if only moderate in strength. In other words, inference gave the clearest explanatory insight, but prediction gave the more actionable result. For this reason, prediction is more useful for the main problem of credit-risk screening, while inference remains essential for interpreting the drivers of default and explaining the model's conclusions.

10. Conclusion

The report examined credit default risk inferentially and predictively using data from 30,000 Taiwanese credit card holders. Our analysis identified repayment delay status (PAY_0) as the most significant predictor of default with a coefficient estimate of 0.65 across frequentist, Bayesian, and bootstrap approaches. Predictive modeling demonstrated that while logistic regression achieves moderate discrimination (PR-AUC ≈ 0.5), class balancing meaningfully improves default detection by raising recall from 24% to 64%, which is a more relevant outcome for credit risk screening. Regularization experiments further confirmed that the dataset's signal is broadly distributed, making the unpenalized model a reliable baseline. To conclude, inference reveals the interpretable drivers of default behavior, while prediction provides the actionable classifications that inform real-world lending decisions.